Optimizing Temporal Regularity of Noise-Induced Spikes in the FitzHugh-Nagumo Model

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1 Introduction

Neurons communicate through electrical signals known as action potentials, which arise from complex interactions between ionic currents across the cell membrane. The classical model describing this behavior is the Hodgkin–Huxley model, which accurately captures the dynamics of ion channels responsible for spike generation. However, the Hodgkin-Huxley model consists of four coupled nonlinear differential equations with many parameters to consider, making it extremely difficult to conduct analytical studies utilizing the model.

To better understand the qualitative mechanism underlying neuronal excitability, we use the FitzHugh-Nagumo model which still captures the essential dynamics of spike generation while only using two variables: one representing membrane potential and another representing inhibitory processes. And despite the reductions in the model, the FitzHugh-Nagumo model still reproduces key behaviors such as threshold dynamics, refractory periods, and oscillatory spiking.

In ideal, deterministic settings, the FitzHugh-Nagumo system exhibits dynamics that can be studied using analytical tools from nonlinear dynamical systems theory like phase portraits, nullclines, and limit cycle analysis. For example, the Poincaré-Bendixson theorem allows us to determine if there exists a limit cycle (periodic spiking behavior) within some trapping region in a phase space. However, real biological systems are inherently noisy. Ion collisions, thermal fluctuations, and other phenomenon introduce randomness into neuronal dynamics. To account for these effects, we turn to the stochastic Fitz-Hugh-Nagumo system, typically formulated as a system of SDEs.

The goal of this project is to investigate the dynamics of the stochastic FitzHugh-Nagumo model using both theoretical analysis and numerical simulations. In particular, we explore how stochastic perturbations affect the regularity of spike trains and modify the long-term dynamics compared to the deterministic case. By combining analytical methods with computational experiments, we illustrate how noise interacts with nonlinear dynamics in excitable biological systems.

2 Research Question and Hypothesis

The central research question of this project is: Is there an optimal noise intensity that maximizes the temporal regularity of noise-induced spikes in the excitable FitzHugh-Nagumo model? In other words, can action potentials achieve near-periodic behavior in the presence of stochastic perturbations, given that the system is excitable (membrane potential stays at resting potential after initial spike)? Temporal regularity refers to the degree to which successive spikes occur at approximately equal time intervals. If the noise is too weak, spikes may occur very rarely and at irregular times. Conversely, if the noise is too strong, fluctuations may dominate the dynamics and produce highly irregular spiking behavior. Based on this reasoning, we hypothesize that there exists an intermediate noise intensity that produces spikes occurring at approximately equal time intervals, resulting in maximized temporal regularity of noise-induced spiking. This phenomenon is related to the concept of coherence resonance, in which stochastic perturbations can enhance the regularity of oscillations in excitable systems.

3 Model Formulation and Simulation

3.1 Deterministic FHN

For the deterministic model, we study the set of equations

$$\dot{v} = v - \frac{v^3}{3} - w + I \tag{1}$$

$$\dot{w} = \varepsilon(v + a - bw) \tag{2}$$

where v represents membrane potential, w represents a recovery/inhibitory process, I represents stimulus current, a represents resting membrane potential, b represents strength of recovery, and ε represents a time-scale constant. Following convention set by FitzHugh (1961), we set our parameters as $a = 0.7$, $b = 0.8$, and $\varepsilon = 0.08$, putting our system in an excitable regime at small I.

The nullclines of (1) and (2) are

$$w = v - \frac{v^3}{3} + I \tag{3}$$

$$w = \frac{v+a}{b} \tag{4}$$

, respectively. Thus, solving for the stationary points would yield different results depending on the input current. For example, when $I = 0.1$, there is one stationary point at approximately $(-1.14, -0.55)$. When $I = 0.5$, there is one stationary point at approximately $(-0.80, -0.13)$. When $I = 0.9$, one stationary point at approximately $(0.10, 1.0)$. These values of I are chosen to fall within a biological range of $I \in [0, 1.5]$ following the parameter study conducted by Hanan et al. (2020). The stability of these points can be determined by analyzing the Jacobian matrix of the system at each point. Consider the Jacobian matrix:

$$J = \begin{bmatrix} 1 - v^2 & -1 \\ \varepsilon & -\varepsilon b \end{bmatrix} \tag{5}$$

We compute the eigenvalues of J at each stationary point to determine their stability. For $I = 0.1$, the eigenvalues are complex conjugates with negative real parts, indicating that the stationary point is a stable focus. In this regime, the system is excitable such that trajectories remain near the equilibrium under small perturbations, but sufficiently large perturbations can trigger a spike (action potential) before returning to rest. For $I = 0.5$, the eigenvalues are complex conjugates with positive real parts, indicating that the stationary point is an unstable focus. Similarly, for I=0.9, the eigenvalues are real and positive, corresponding to an unstable node. In both cases, the equilibrium has lost stability through a Hopf bifurcation (at some $I = I_H$). But we still don't know what the long-term behavior of the system is for $I > I_H$. Does the membrane potential diverge to infinity? From a biological standpoint, probably not. Does it approach a limit cycle? Maybe. This is more likely but we'll need to prove this somehow. We can use the Poincaré-Bendixson theorem to help us analyze long-term behavior here. The theorem states that if a trajectory of a two-dimensional dynamical system is confined to a compact region and does not converge to a stationary point, then it must approach a limit cycle. In other words, since we already know that at $I = 0.5, 0.9$ the stationary point is unstable, if we can show that trajectories are bounded and do not diverge to infinity, then we can conclude that there exists a limit cycle for $I > I_H$ which corresponds to periodic spiking behavior in the neuron.

Although we don't exactly have a trapping region that we can explicitly define, a good first step is to show that the divergence of the vector field is negative which would imply that the system is dissipative and suggest that trajectories may be confined to some bounded region in the phase space. The divergence of the vector field is given by

$$\nabla \cdot F = \frac{\partial \dot{v}}{\partial v} + \frac{\partial \dot{w}}{\partial w} = (1 - v^2) - \varepsilon b \tag{6}$$

Since εb is a positive constant, the divergence will be negative for sufficiently large values of v. However, this still isn't enough yet. We can also analyze the vector fields to gain insight into the behavior of trajectories. We see that to leading order, $\dot{v}$ is dominated by the $-v^3/3$ term for large $|v|$, which will cause trajectories to be pushed back towards the origin in the v-direction when $|v|$ is large. We can split this in two cases:

1. For $v > 0$ and large in magnitude, the cubic nullcline will be negative and thus push trajectories downwards.

2. For $v < 0$ and large in magnitude, the cubic nullcline will be positive and thus push trajectories upwards.

So combined with the dissipativity argument, this loosely suggests that we may have a trapping region in phase space and since at $I > I_H$ the stationary point is unstable, we can hypothesize that there exists a limit cycle for $I > I_H$ which corresponds to periodic spiking behavior in the neuron. To test this claim, we can run various simulations and study the plots generated. Firstly, we can plot the phase portraits of the system for each value of I. The phase portraits show the trajectories of the system in the v-w plane, along with the nullclines and stationary points.

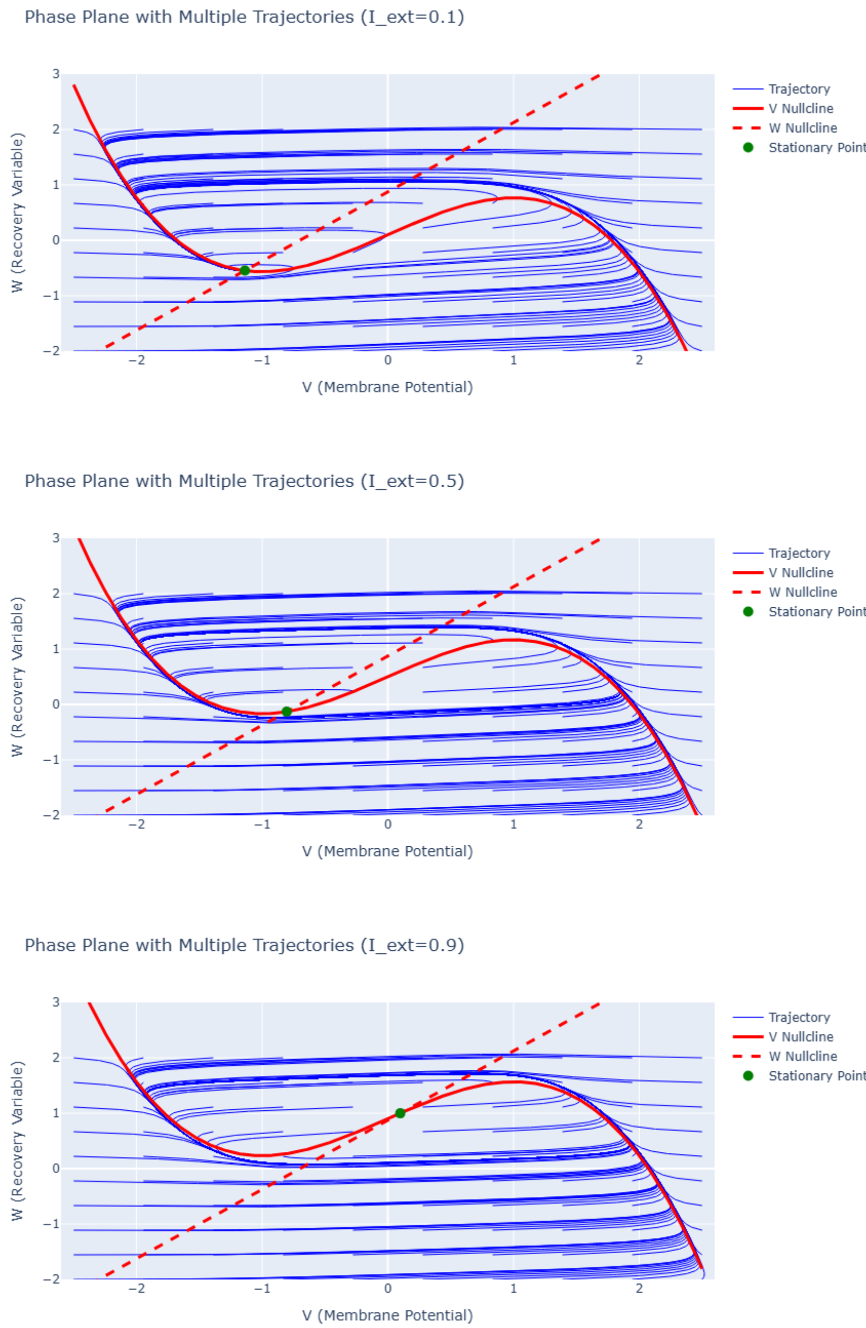
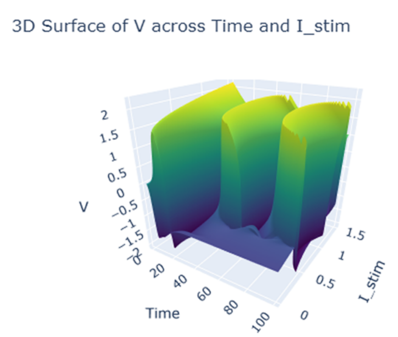


Figure 1: Phase portraits of the deterministic FitzHugh-Nagumo model for different values of input current I. The nullclines are shown in red lines and the stationary points are marked with dots. The trajectories illustrate the dynamics of the system in the v-w plane.

For $I = 0.1$, we observe that trajectories spiral towards the stable focus, indicating that the system is in an excitable regime. For $I = 0.5$, we observe that trajectories spiral around the unstable focus. And similarly, for $I = 0.9$ we see that trajectories also never settle at the stationary point. These phase portraits provide insight into the qualitative dynamics of the system under different input currents and essentially confirm our previous analysis of the stability of the stationary point.

To gain a further understanding of how the system evolves over time, we can also plot the time series of v and w for each value of I. The time series show how the membrane potential and recovery variable change over time, providing insight into the temporal dynamics of the system. What we see is that from $I = 0$ to $I = 0.3$, the system stays at resting membrane potential after the initial spike. However, for $I > 0.3$, the system exhibits unstable to periodic spiking behavior which confirms our loose claims about the system's oscillatory behavior at $I > I_H$.



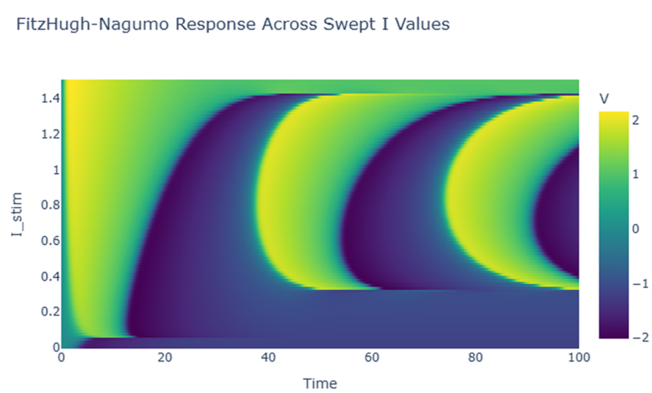


Figure 2: Time series of the membrane potential v for different values of input current I. The plots illustrate how the system evolves over time, showing the transition from resting state to periodic spiking behavior as I increases from 0 to 1.5.

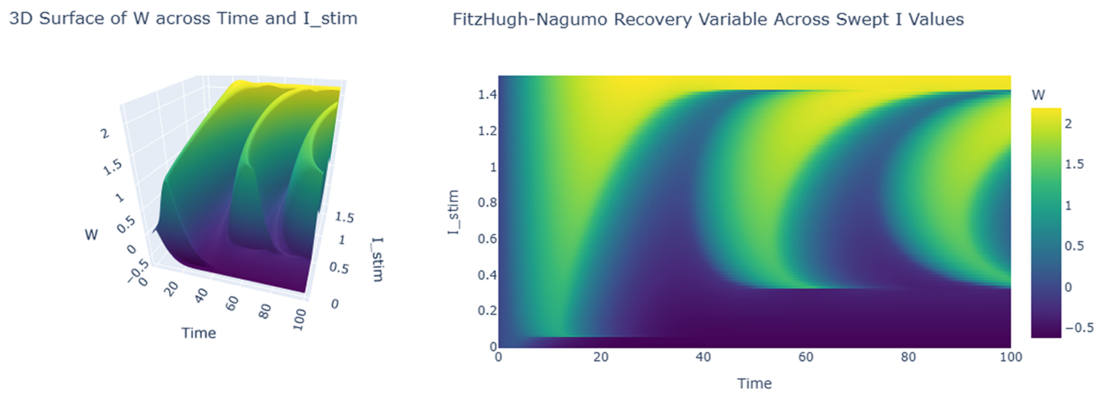


Figure 3: Time series of the recovery variable w for different values of input current I.

The deterministic system therefore exhibits two qualitatively unique regimes of behavior depending on the value of I: an excitable regime for low input currents and an oscillatory regime for medium to high input currents. The excitable regime is particularly important for our study because it is in this regime that noise can induce spikes. In real neurons, such perturbations often arise from random fluctuations caused by ion channel noise or other stochastic processes, thus motivating our extension of the deterministic formulation to a stochastic version of the model.

3.2 Stochastic FHN

For the stochastic model, we study the set of equations

$$dv = \left(v - \frac{v^3}{3} - w + I\right) dt + \sigma dW_t \tag{7}$$

$$dw = \varepsilon(v + a - bw)dt \tag{8}$$

where σ represents the volatility or noise intensity and W_t represents a standard Wiener process. Notice that when we take $\sigma = 0$, our system mirrors the deterministic case. The noise term σdW_t introduces stochastic perturbations to the membrane potential, which can lead to noise-induced spiking behavior even when the system is in a resting state in the deterministic case.

To investigate the effect of noise on the dynamics of the system, we can perform numerical simulations of the stochastic FitzHugh-Nagumo model for different values of σ. Most commonly, the Euler-Maruyama method is used to approximate the solution of the SDEs, which involves discretizing time into small steps and iteratively updating the values of v and w. Thus, we will use

$$v_{n+1} = v_n + \left(v_n - \frac{v_n^3}{3} - w_n + I \right) \Delta t + \sigma \sqrt{\Delta t} Z_n \tag{9}$$

where we have approximated the increment ΔW_n as $\sqrt{\Delta t} Z_n$ based on the properties of Brownian motion and used Z_n as standard normal random variable $N(0, 1)$. We can also update w using the deterministically since it does not have a noise term:

$$w_{n+1} = w_n + \varepsilon (v_n + a - b w_n) \Delta t \tag{10}$$

Literature suggests that $\sigma = 0.2$ is considered strong noise so we will sweep σ from 0.0 to 0.4 to stay within a physiologically relevant range. Additionally, we set $I = 0.1$ so that the deterministic system remains at a stable resting equilibrium (excitable regime), allowing us to isolate noise-induced spikes and test whether an intermediate noise intensity maximizes their temporal regularity (coherence resonance).

For each value of σ, we will run 60 trials of the simulation to compute statistics on the inter-spike intervals (ISIs) and spike counts, which will allow us to analyze the temporal regularity of the noise-induced spikes. Then we can plot the mean coefficient of variation (CV) of the ISIs and the mean spike count as a function of σ to investigate whether there is an optimal noise intensity that maximizes temporal regularity. Note that we define the CV of ISIs as

$$CV = \frac{\sigma_{ISI}}{\mu_{ISI}} \tag{11}$$

where σ_{ISI} is the standard deviation of the ISIs and μ_{ISI} is the mean of the ISIs. A lower CV indicates more regular spiking behavior, while a higher CV indicates more irregular spiking behavior. Therefore, if there exists an optimal noise intensity that maximizes temporal regularity, we would expect to see a minimum in the mean CV as a function of σ, the noise intensity.

We now examine the results of the stochastic simulations across the range of noise intensities. For each value of σ, the model was simulated for multiple trials and the resulting spike statistics were aggregated to compute the mean coefficient of variation (CV) of the inter-spike intervals and the mean spike count.

Figure 4 shows the mean CV of the inter-spike intervals as a function of the noise intensity. For very small values of σ, the system rarely produces spikes because the deterministic system lies in a resting state. As a result, the few spikes that do occur are highly irregular, leading to large variability in the CV. As the noise intensity increases, the system begins to spike more regularly due to noise-induced threshold crossings.

Interestingly, the CV initially decreases as σ increases, indicating that the spike timing becomes more regular. This phenomenon is somewhat related to coherence resonance but we ultimately don't observe a clear minimum in the mean CV as a function of σ. Instead, the CV appears to plateau at intermediate to high noise intensities, suggesting that while noise can enhance the regularity of spiking to some extent, there may not be a well-defined optimal noise intensity that maximizes temporal regularity in this particular system.

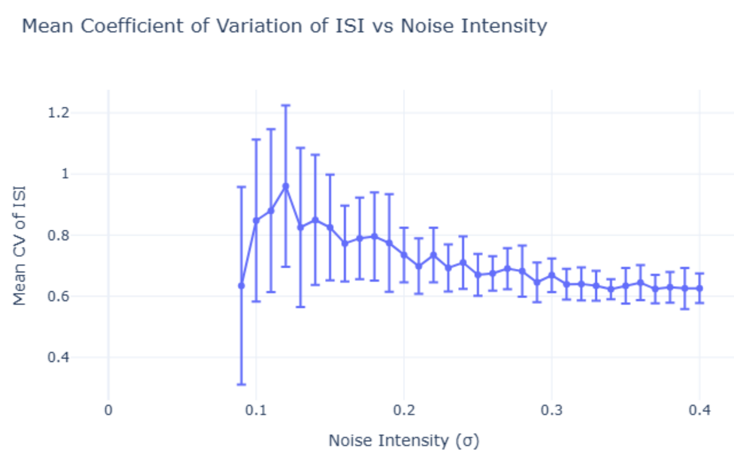


Figure 4: Mean coefficient of variation (CV) of the inter-spike intervals as a function of noise intensity σ. Error bars represent the standard deviation across trials.

To further understand the effect of noise on the activity of the system, we also examine the mean spike count as a function of noise intensity. The results are shown in Figure 5. As expected, the number of spikes increases roughly monotonically with increasing σ. When σ is small, the noise is insufficient to push the membrane potential past the excitation threshold, resulting in few or no spikes. As the noise intensity increases, stochastic fluctuations increasingly drive the system across the threshold, leading to more frequent spiking events.

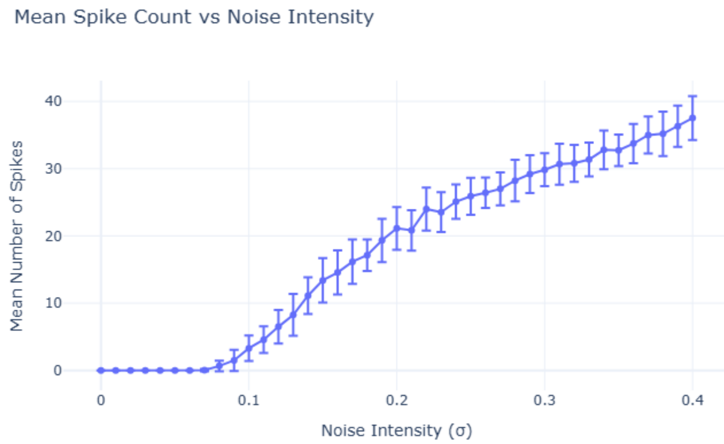


Figure 5: Mean spike count as a function of noise intensity σ. Error bars represent the standard deviation across trials.

Finally, to illustrate how the qualitative behavior of the membrane potential changes with noise intensity, sample trajectories of the membrane potential $v(t)$ are shown for several representative values of σ in Figure 6. At very low noise levels, the system remains near its resting state with only small fluctuations. As σ increases, occasional excursions away from the resting state occur, producing isolated spikes. For larger values of σ, the system exhibits sustained stochastic spiking with increasingly frequent firing events.

These trajectories highlight how noise can fundamentally alter the dynamical behavior of an excitable system. Even though the deterministic model remains at rest, stochastic perturbations can induce repeated threshold crossings and generate spike trains.

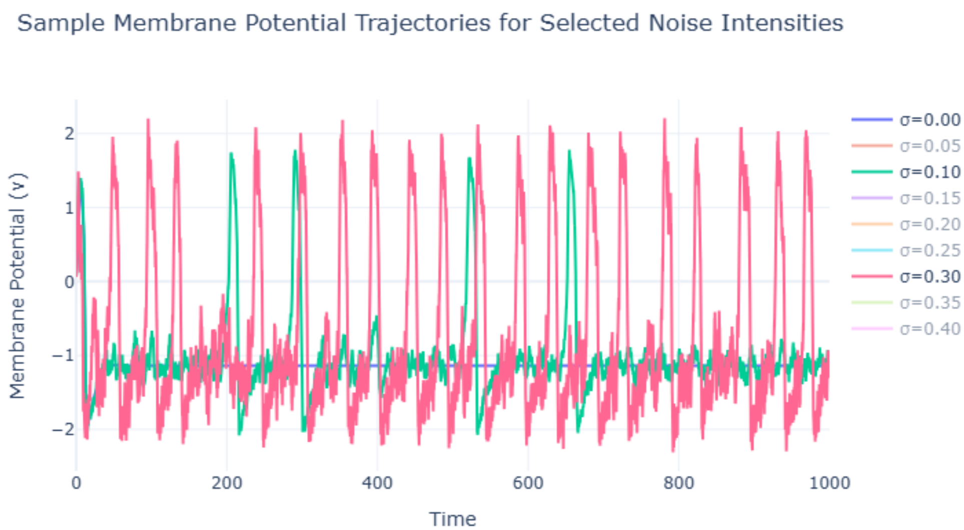


Figure 6: Sample membrane potential trajectories $v(t)$ for selected values of the noise intensity σ. Increasing noise leads to more frequent noise-induced spikes.

Together, these results demonstrate how stochastic perturbations can induce spiking behavior in an excitable system and influence the temporal regularity of the resulting spike trains.